

CS4234: Optimization Algorithms
Semester 1, AY 2026–27
School of Computing, NUS

Problem Set for Tutorial 3

Question 1: A Randomized Approximation for MAX k -CUT

Let $G = (V, E)$ be an undirected graph with nonnegative edge weights w_{ij} , and let $k \geq 2$. A feasible solution partitions V into k possibly empty parts $V_1, \dots, V_k$. Its value is the total weight of the edges whose endpoints lie in different parts.

Give a randomized polynomial-time algorithm whose expected value is at least $\frac{k-1}{k}\text{OPT}$. Prove the approximation guarantee.

Question 2: A Basic Randomized Approximation for MAX DICUT

Let $G = (V, A)$ be a directed graph with nonnegative arc weights w_{ij} . Assume that G has no self-loops; such loops can be discarded because they never cross a directed cut. A directed cut is a partition (U, W) of V , where $W = V \setminus U$. Its value is the total weight of the arcs $(i, j) \in A$ with $i \in U$ and $j \in W$.

Give a randomized polynomial-time algorithm whose expected value is at least $\frac{1}{4}\text{OPT}$. Prove the approximation guarantee.

Question 3: LP Rounding for MAX DICUT

Use the same notation as in Question 2.

3A. An integer-programming formulation

For each vertex $i \in V$, let x_i indicate whether i is placed in U . For each arc $(i, j) \in A$, let z_{ij} indicate whether that arc contributes to the directed cut. Show that the following integer program

has an optimum equal to the maximum directed-cut value:

$$\begin{aligned}
& \text{maximize} && \sum_{(i,j) \in A} w_{ij} z_{ij} \\
& \text{subject to} && z_{ij} \leq x_i, && \text{for every } (i,j) \in A, \\
& && z_{ij} \leq 1 - x_j, && \text{for every } (i,j) \in A, \\
& && x_i \in \{0,1\}, && \text{for every } i \in V, \\
& && 0 \leq z_{ij} \leq 1, && \text{for every } (i,j) \in A.
\end{aligned}$$

Explain both directions: how a directed cut induces a feasible solution of the same value, and how an optimal integral solution induces a directed cut of the same value.

3B. Randomized rounding

Relax the constraints $x_i \in \{0,1\}$ to $0 \leq x_i \leq 1$ and solve the resulting linear program. Let (x,z) be an optimal LP solution. Independently place each vertex i in U with probability

$$p_i = \frac{1}{4} + \frac{x_i}{2},$$

and place every remaining vertex in W .

Prove that the expected weight of the resulting directed cut is at least half the LP optimum, and hence that this is a randomized 2-approximation for MAX DICUT.